

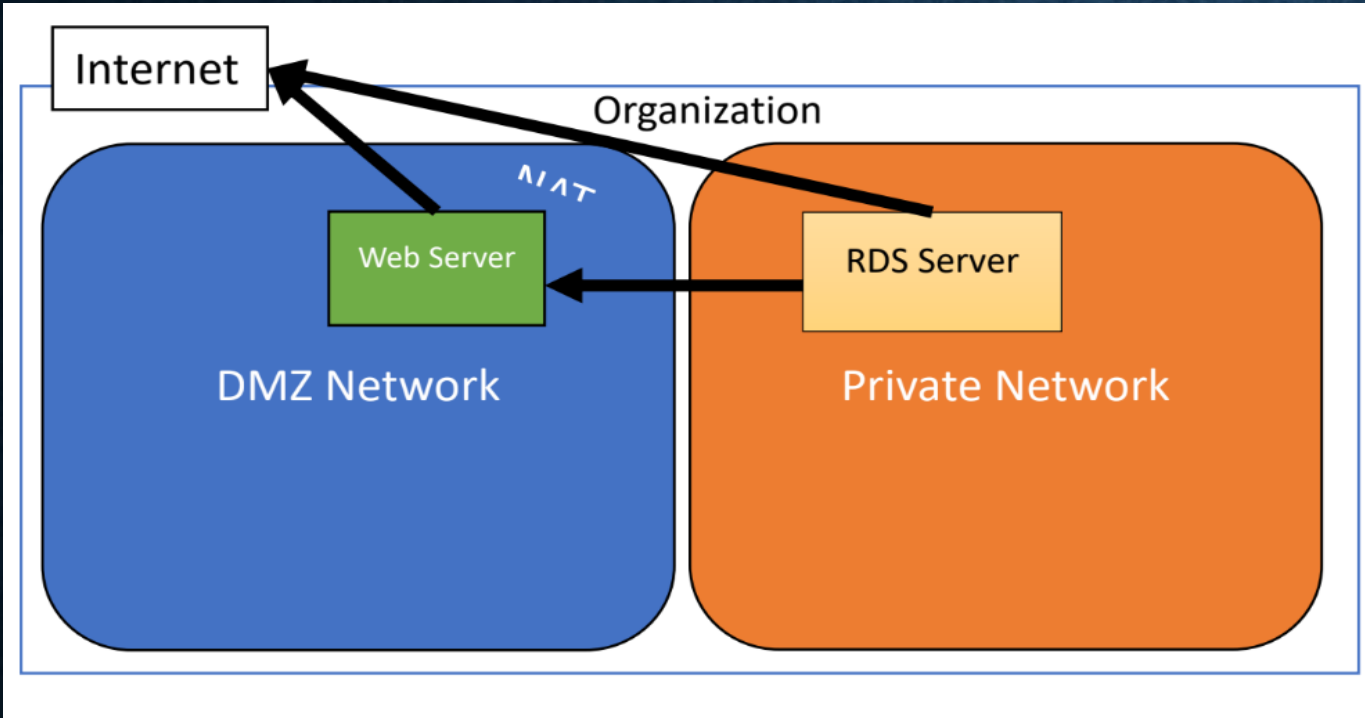


Name: *Sayantan Das*

Batch: **Cloud computing with AWS**
(Jul-Aug 2024)

Project: **Secured Web Server Implementation**

PROJECT OVERVIEW



- Two ec2 instances will be created which will be attached to the respective subnets.
- Then I will install & configure the given servers to the respective virtual machines.
- Lastly I will create a load balancer to support and display both the servers simultaneously.

- To implement the project, I am going to use my account logged in to the AWS console and use its free-tier resources.
- With reference to the image, I am going to use the AWS resources.
- To build the organization I am going to create my own VPC which will be connected to internet gateway created by me.
- I am going to create two different subnets of different zones for contrasting networks which will also be routed to the same internet gateway.

****Please do zoom in the images if not properly visible as it is.****

Let's begin!!

STEP 1: CREATE VPC

aws

Services

Q

Search

[Alt+S]

Mumbai

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Create VPC

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

my_vpc

IPv4 CIDR block

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

192.168.0.0/16

CIDR block size must be between /16 and /28.

CloudShell

Feedback

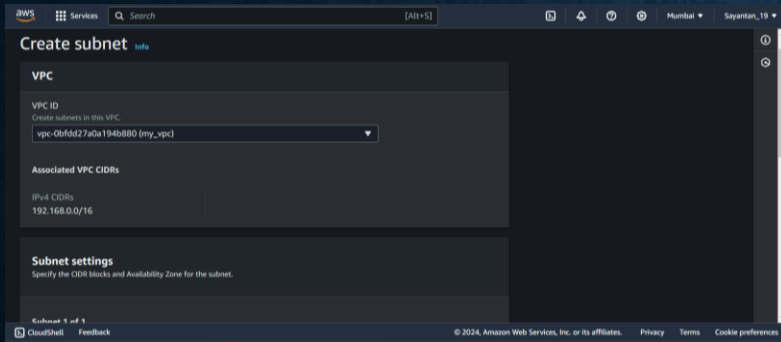
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STEP 2: CREATE SUBNETS



Create subnet [info](#)

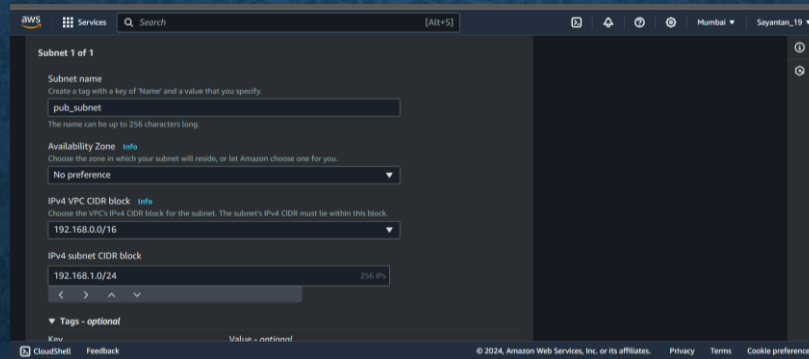
VPC

VPC ID
Create subnets in this VPC.
vpc-0b6d27a0a194b880 (my_vpc)

Associated VPC CIDRs

IPv4 CIDRs
192.168.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.



Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
pub_subnet
The name can be up to 256 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must be within this block.
192.168.0.0/16

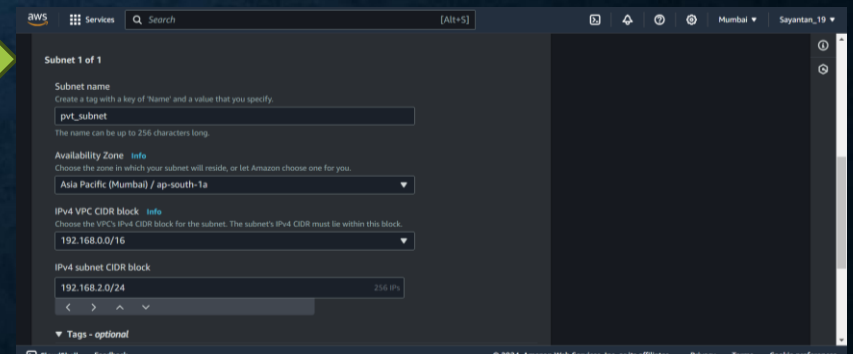
IPv4 subnet CIDR block
192.168.1.0/24 256 IPs

Tags - optional

Creating pub_subnet for public network. (got ap-south-1b)



Creating pvt_subnet for private network.



Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
pvt_subnet
The name can be up to 256 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must be within this block.
192.168.0.0/16

IPv4 subnet CIDR block
192.168.2.0/24 256 IPs

Tags - optional

STEP 3: CREATE INTERNET GATEWAY

aws

Services

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Create internet gateway

Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

my_internet

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q my_internet X

Remove

Add new tag

You can add 49 more tags.

CloudShell

Feedback

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The following internet gateway was created: igw-03e8f29b937292683 - my_internet. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to a VPC

VPC

Internet gateways

Attach to VPC (igw-03e8f29b937292683)

Attach to VPC (igw-03e8f29b937292683)

Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

Q vpc-0bfdd27a0a194b880 X

AWS Command Line Interface command

Cancel

Attach internet gateway

CloudShell

Feedback

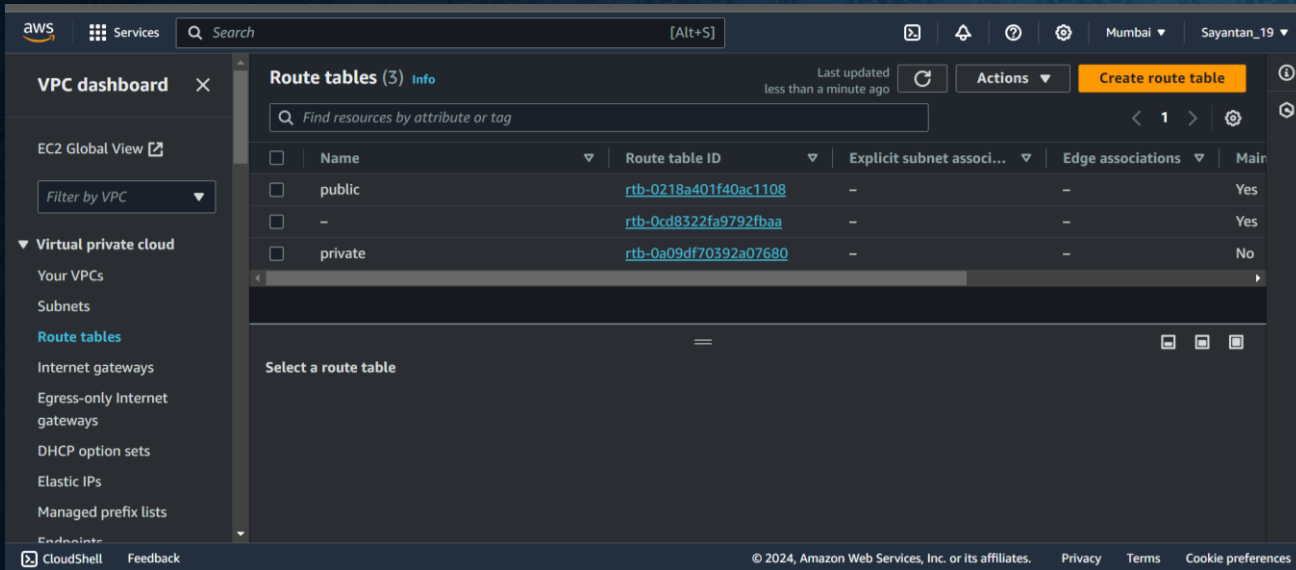
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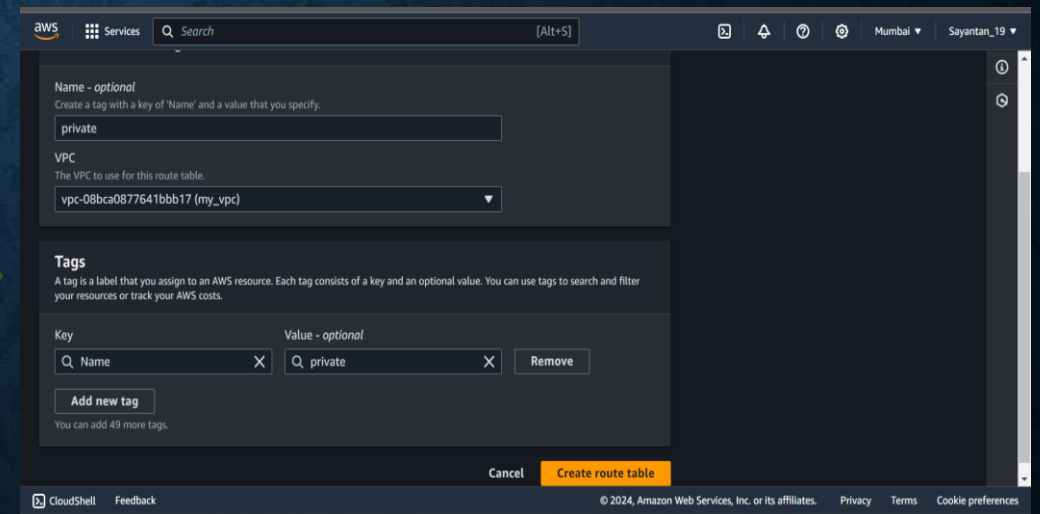
Cookie preferences

STEP 4: CREATE ROUTE TABLES

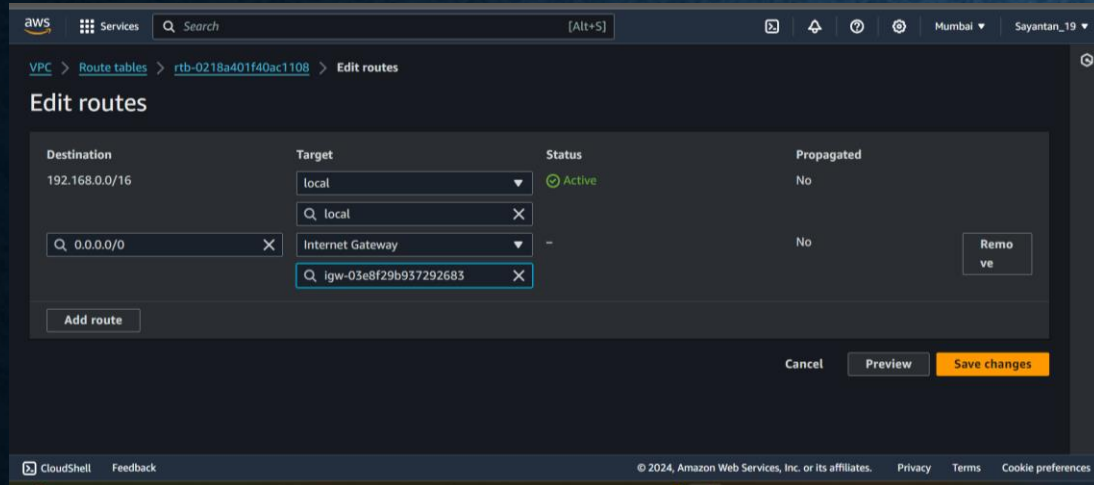


The route table which was created by default while creating “my_vpc” is renamed as public which will be used for public routing.

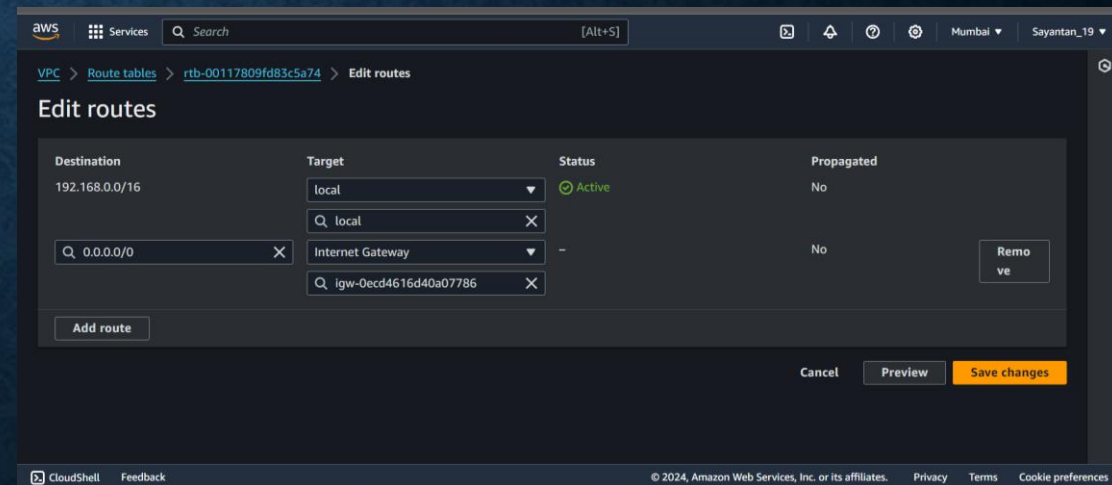
Route table for private routing is separately created and named as private.



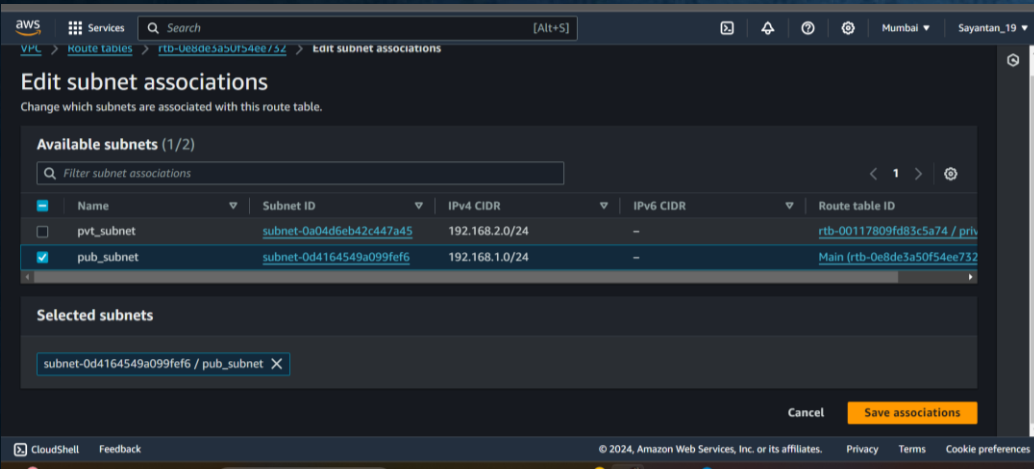
STEP 5: EDIT ROUTES



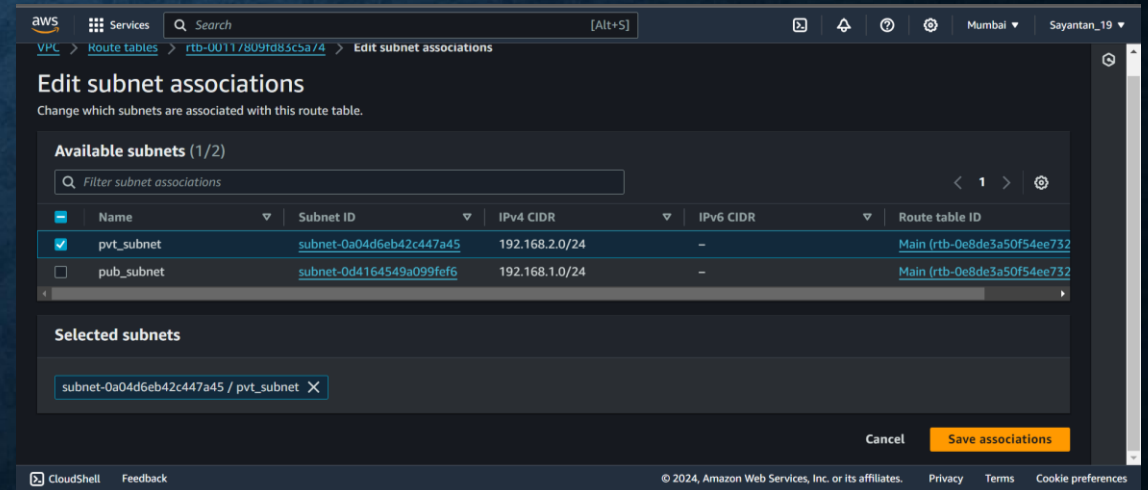
Editing routes for both subnets to be connected to the common internet gateway “my_internet”.



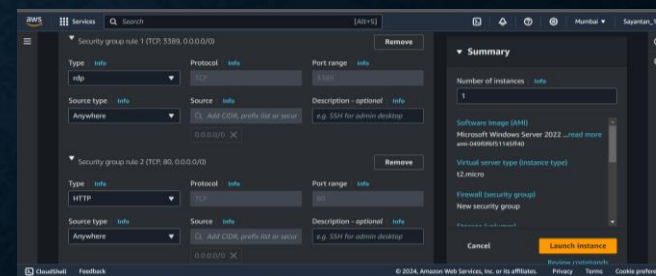
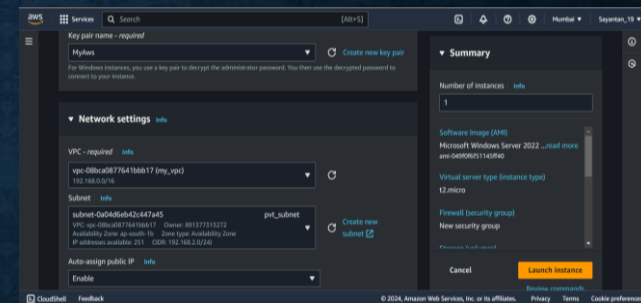
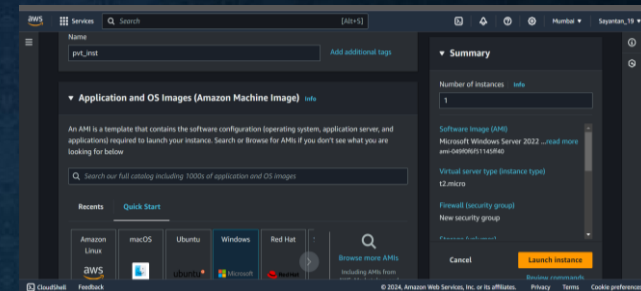
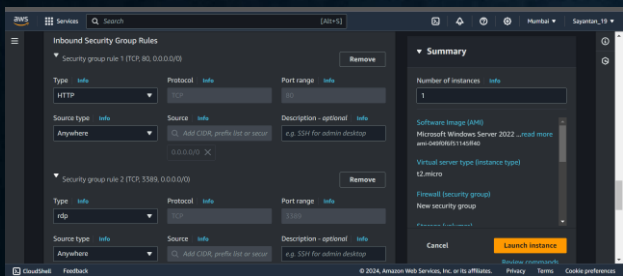
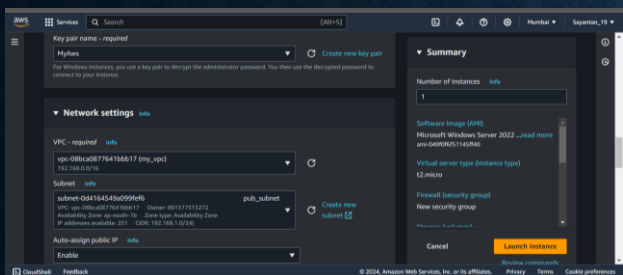
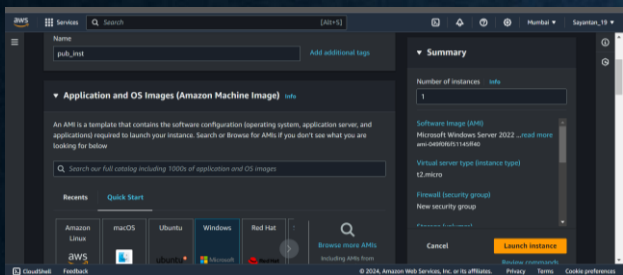
STEP 6: EDIT SUBNET ASSOCIATIONS



Associating public subnet and private subnet to public route and private route respectively.



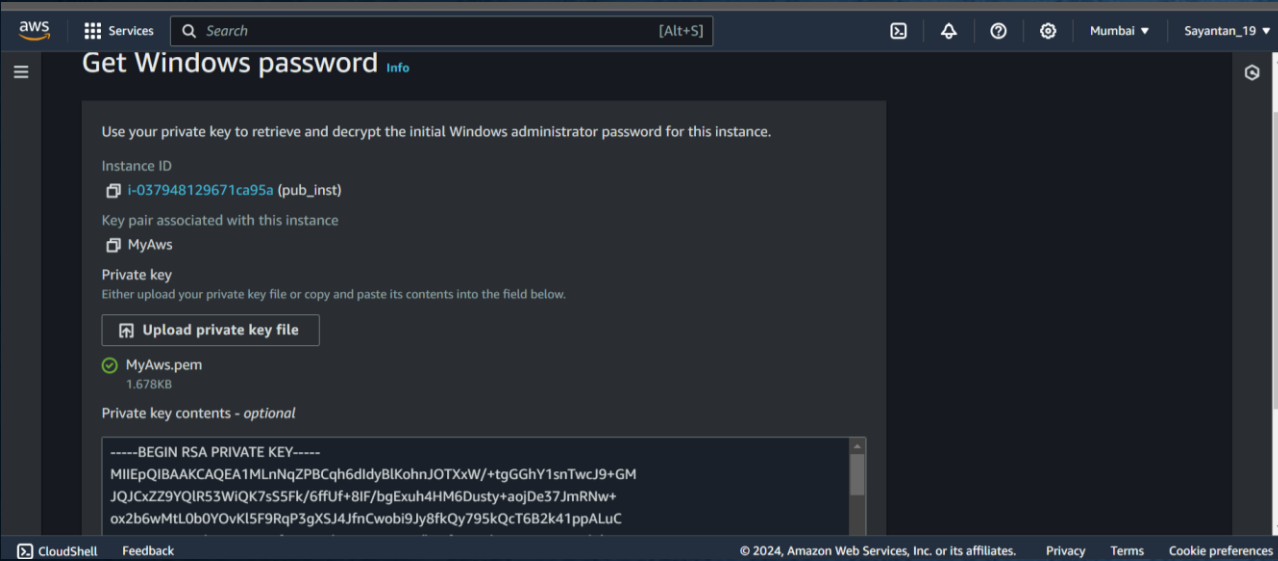
STEP 7: LAUNCH EC2 INSTANCES



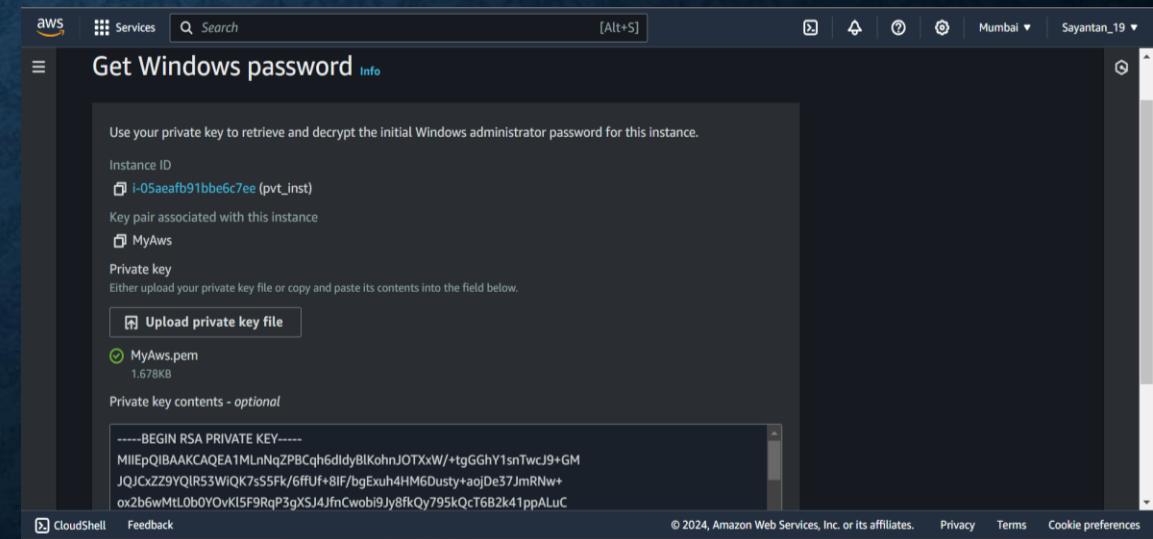
Two ec2 instances is attached to public subnet and private subnet respectively.

Allowing type: HTTP & rdp security inbound rules to both the instances.

STEP 8: OPEN BOTH THE VIRTUAL MACHINES



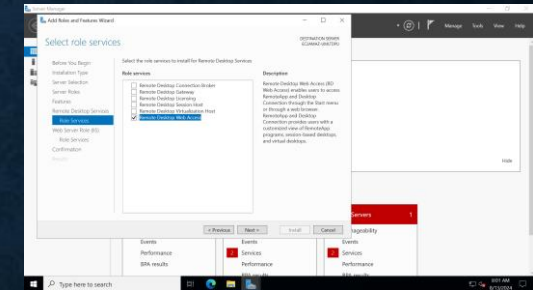
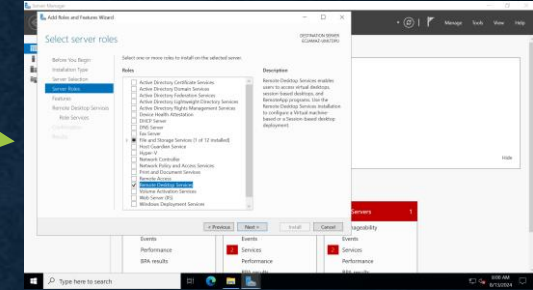
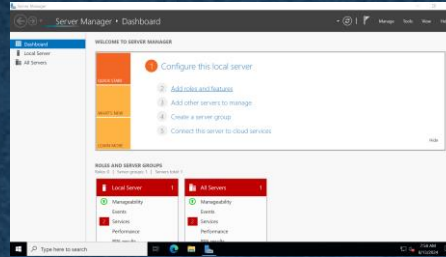
Decrypting the password using
my key pair and using the same to
open both the virtual machines.



STEP 9: INSTALL & CONFIGURE SERVERS

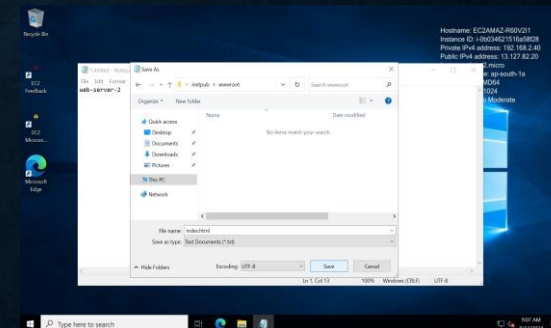
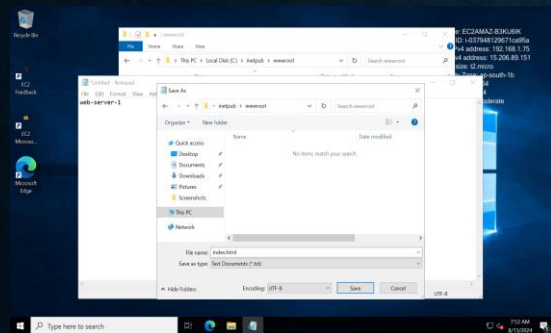
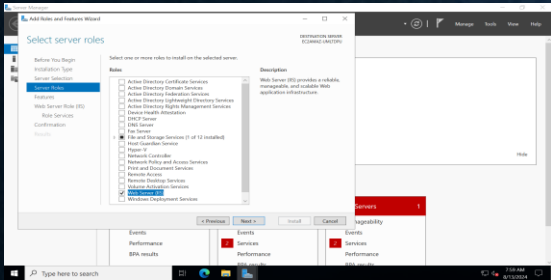
Configuring Web Server in this instance(pub_inst) for publicly accessible networking(DMZ).

Server manager opened in both virtual machine.

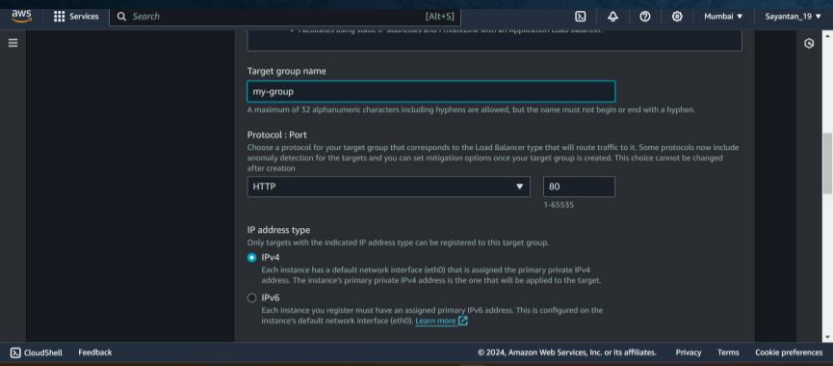


Configuring RDS Server with Web Server(for web access) in this instance(pvt_inst) for private networking.

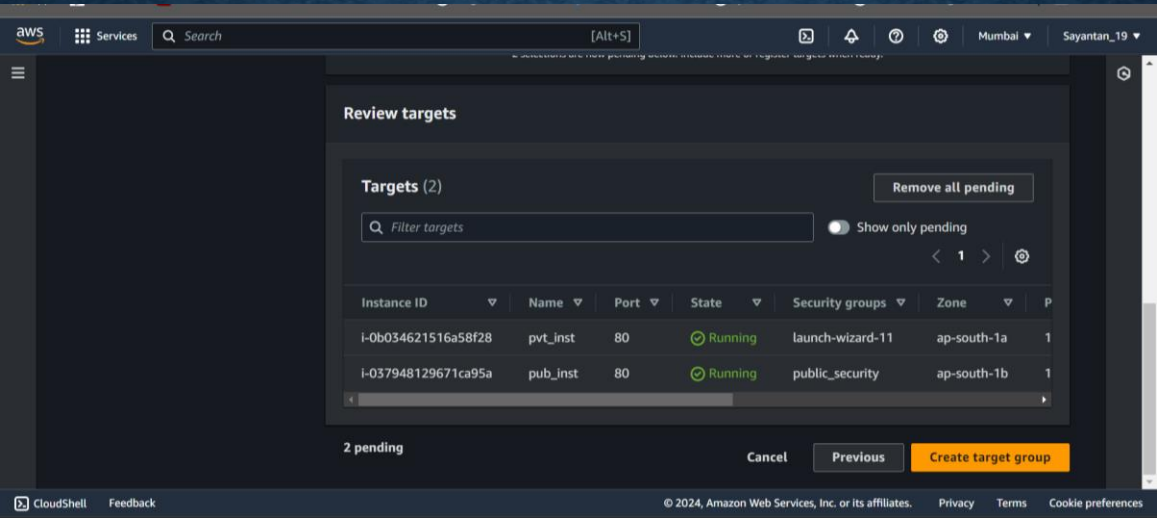
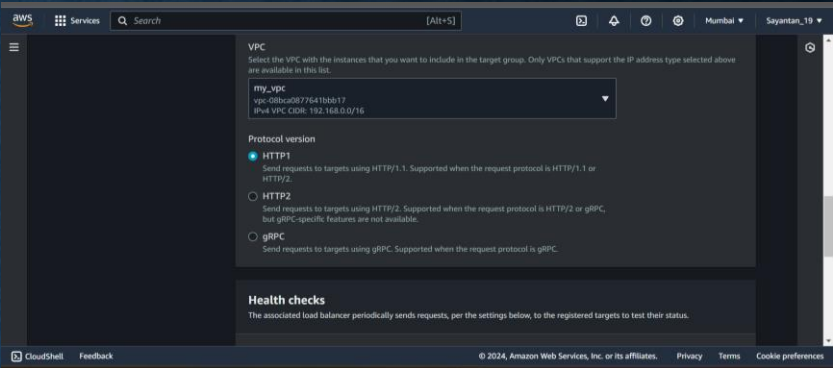
****An html file is created and saved in both the virtual machines in order to check the working of both the servers.****



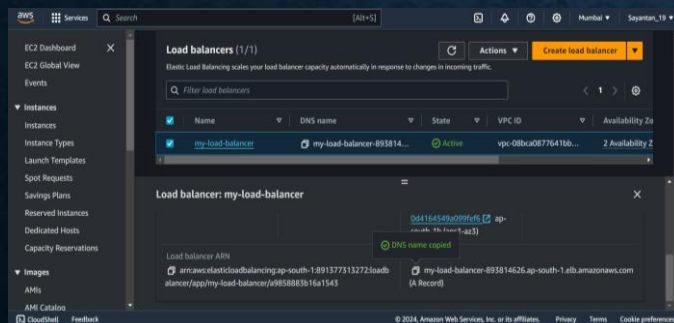
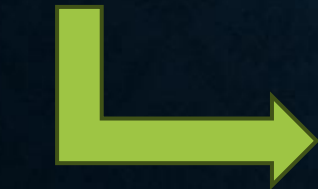
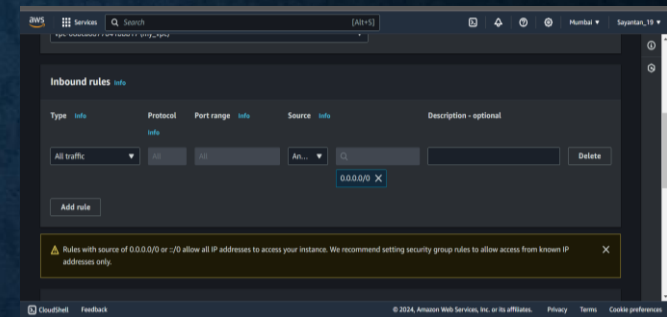
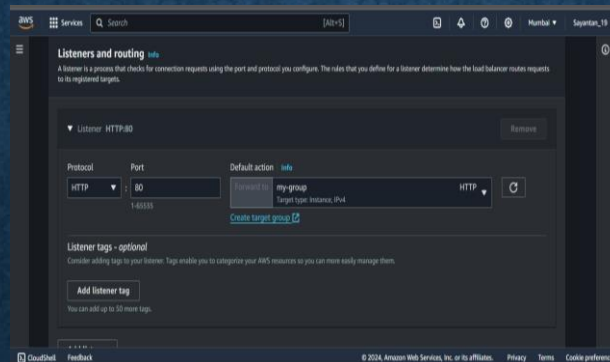
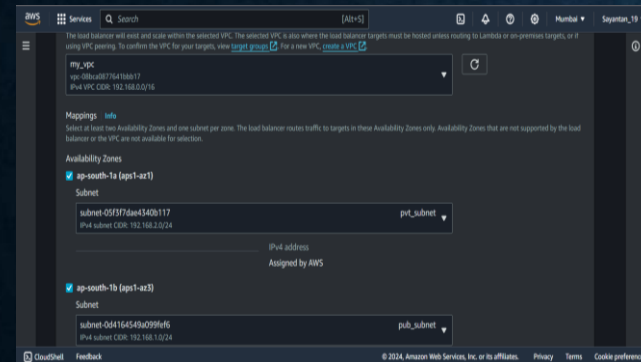
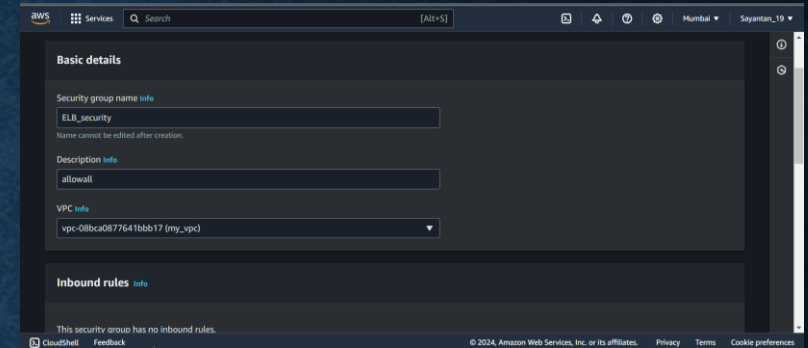
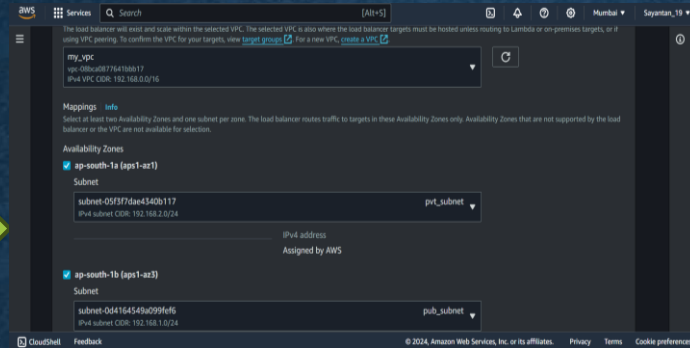
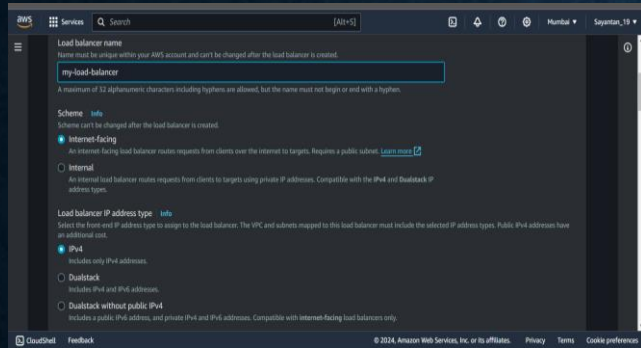
STEP 10: CREATE TARGET GROUP



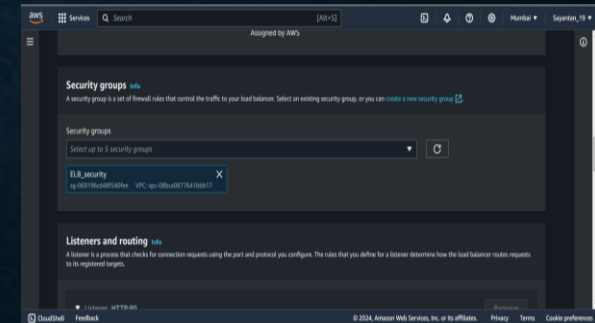
Both the ec2s are kept in the target group named “my-group”.



STEP 11: CREATE APPLICATION LOAD BALANCER

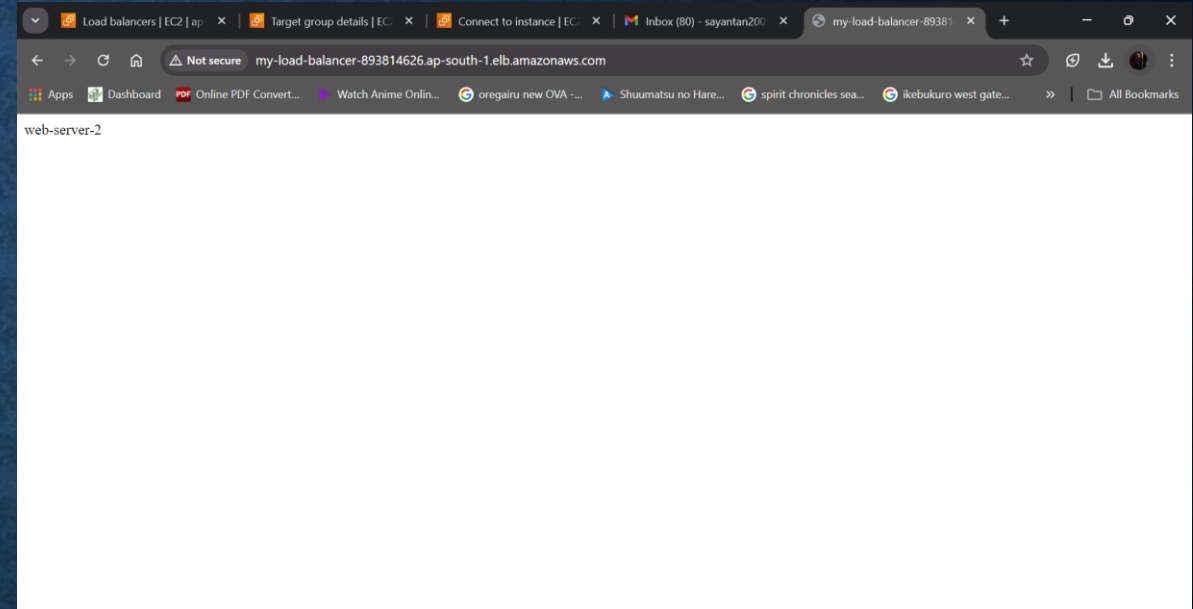
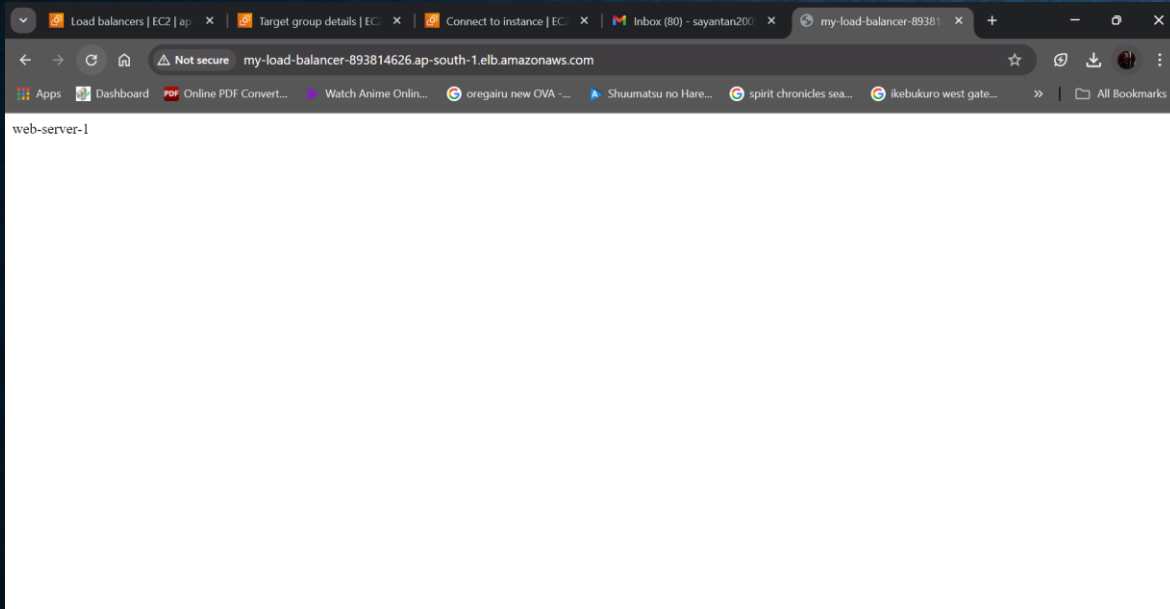


Creating separate security group allowing “all traffic” in load balancer.



STEP 12: RESULT

After the activation of load balancer → Copy DNS name of it → Paste it in new tab



As shown in the images above, the file which we configured in the virtual machine is getting displayed successfully and due to the load balancer we can access both the files from the respective virtual machines simultaneously using the reload button.



Thank
you!!